Cuneus Grey Matter Volume and Picture Vocabulary Performance After Preterm Birth

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Developed from the poster “Cuneus Cortex Mediates Lower Cognitive Performance in Adolescents Born Premature” (Arya Shah, Dawn Jensen, Vince Calhoun, Jingyu Liu). Poster venue and date [AUTHOR TO CONFIRM].

Abstract

Background. Preterm birth is associated with later differences in brain development and cognition, but less is known about how these differences relate to each other in adolescence.

Aim. To test whether regional grey matter (GM) volume helps account for lower cognitive scores in adolescents born preterm, using data from the Adolescent Brain Cognitive Development (ABCD) Study.

Methods. Adolescents were grouped as full-term (n = 2,252), moderately premature (32–37 weeks; n = 731) or very premature (<32 weeks; n = 152). GM volume was estimated for 100 regions from structural MRI using moo-ICAr. Linear mixed-effects models with sex and gestational age as fixed effects and site as a random effect compared GM and NIH Toolbox scores across groups. A mediation model tested gestational age → cuneus GM volume → Picture Vocabulary score.

Results. Five GM regions differed by group. Cuneus GM volume was higher in moderately premature (t = 4.36) and very premature (t = 3.84) adolescents than in full-term adolescents. Picture Vocabulary scores were lower in moderately premature (t = −3.13) and very premature (t = −5.59) adolescents. Cuneus GM volume accounted for 4.1% of the association between gestational age and Picture Vocabulary performance.

Conclusions. In this sample, preterm birth was associated with higher cuneus GM volume and lower receptive vocabulary scores, and cuneus volume statistically mediated a small share of the gestational age–vocabulary association. These are cross-group associations, not evidence of a causal pathway or developmental mechanism. Several methods details await author confirmation before these results can be evaluated.

1. Introduction

Preterm birth, defined as birth before 37 completed weeks of gestation, affected an estimated 13.4 million babies in 2020, more than 1 in 10 births worldwide [1]. Full term is 39 0/7 through 40 6/7 weeks of gestation [2]. Preterm-born school-aged children score lower on cognitive tests than term-born controls on average, and in meta-analysis the size of the gap is proportional to gestational age [3].

Structural MRI studies indicate that these cognitive differences are accompanied by differences in brain structure. In a voxel-based morphometry study of 218 very preterm adolescents (<33 weeks) and 128 controls aged 14–15, very preterm adolescents showed both reduced and increased GM relative to controls in different regions. The reduced GM included the cuneus. Several of these GM and white matter differences were associated with gestational age and statistically mediated cognitive impairment [4]. In a large ABCD analysis, lower gestational age was associated with graded lower cognition and with smaller regional cortical volumes, which mediated part of the gestational age–cognition association [5].

The present study asks a narrower question: whether variation in GM volume helps explain lower cognitive scores in adolescents born preterm in ABCD data. We focus on the cuneus, the region the source poster reports as mediating the association with Picture Vocabulary performance. The poster describes the cuneus as an occipital region involved in visual processing and visual attention [AUTHOR TO ADD CITATION].

Observations and hypotheses. In this draft, an *observation* is a result reported on the source poster. A *hypothesis* is an interpretation not tested by the analysis. Interpretations involving maturation, development over time or visual processing are hypotheses.

2. Methods

2.1 Data source

Data came from the ABCD Study, a 21-site U.S. cohort that recruited over 10,000 children aged 9–10 and follows them through adolescence, with imaging protocols harmonized across sites [6]. The baseline image processing pipeline is described by Hagler et al. [7]. ABCD documentation and derived data are versioned by data release [8]. This draft therefore does not assume that any current release or pipeline describes the data used here.

- Data release and access date: [AUTHOR TO CONFIRM]
- Study visit (timepoint) and participant age range at assessment: [AUTHOR TO CONFIRM]
- Inclusion and exclusion criteria, imaging quality control and missing data handling: [AUTHOR TO CONFIRM]

2.2 Participants and gestational groups

The poster's sample table ("Sample Cohort-1") is reproduced in Table 1. The three rows sum to 3,135 participants. The poster does not state this total or the path from the full ABCD cohort to this sample [AUTHOR TO CONFIRM].

Table 1. Sample by gestational group, as reported on the poster.

Group	Total subjects	Male	Female
Full-term	2,252	1,187 (52.7%)	1,065 (47.3%)
Moderately premature	731	401 (54.9%)	330 (45.1%)
Very premature	152	77 (50.7%)	75 (49.3%)

The poster defines moderately premature as 32–37 weeks and very premature as <32 weeks. Its introduction notes that full term is typically 39–40 weeks. The cutoff actually used for the full-term analysis group is not stated. It is also unclear whether births at 37 and 38 weeks were classed as moderately premature, full-term or excluded [AUTHOR TO CONFIRM]. The source of gestational age (for example, caregiver report) and how it was recorded are also unconfirmed [AUTHOR TO CONFIRM]. The label “Sample Cohort-1” suggests additional cohorts may exist; their role, if any, is unconfirmed [AUTHOR TO CONFIRM].

2.3 Neuroimaging

Structural MRI (sMRI) data were analysed with moo-ICAr to estimate GM volume for 100 regions, as reported on the poster [AUTHOR TO ADD METHOD CITATION]. The following details are not reported on the poster:

- the preprocessing steps (segmentation, normalization, modulation and smoothing)
- the reference template and its origin
- whether the 100 “regions” are independent components or anatomical parcels
- the units of the GM measure

[AUTHOR TO CONFIRM] The poster’s cuneus plot spans roughly −0.1 to 0.5, which suggests a standardized or component-based measure rather than raw volume. Whether models adjusted for intracranial or total brain volume is also unconfirmed [AUTHOR TO CONFIRM].

2.4 Cognition

Cognition was measured with the NIH Toolbox, seven tests including the Picture Vocabulary Test, as reported on the poster. Only Picture Vocabulary results appear on the poster. In the Toolbox Picture Vocabulary Test, participants hear a spoken word while viewing four pictures and choose the picture that best represents the word. The test measures receptive vocabulary without requiring reading or writing [9]. The NIH Toolbox is part of the ABCD neurocognition battery [10]. The score type analysed is unconfirmed: raw, uncorrected standard or age-corrected [AUTHOR TO CONFIRM]. So are results for the other six tests [AUTHOR TO CONFIRM].

2.5 Statistical analysis

The poster describes the following sequence:

1. Linear mixed-effects (LMER) models of GM differences by gestational group. These used fixed effects of sex and gestational age and a random effect of site, written on the poster as `GM ~ group + sex + gestational age`.

2. The same model structure for cognition (`Cognition ~ group + sex + gestational age`).
3. Selection of five significant GM regions.
4. LMER models linking brain structure to cognitive performance.
5. A mediation analysis of the pathway from gestational age (weeks) to Picture Vocabulary score through cuneus GM volume.

Several specification details require confirmation:

- whether group and continuous gestational age entered the same model, given their overlap
- whether age at assessment was a covariate
- whether family or twin relatedness was modelled
- the significance threshold and any correction for testing 100 regions
- why the cuneus was selected among the five regions, and the identity of the other four
- the mediation method, including how the indirect effect was tested and whether a confidence interval was estimated
- the software and package versions

[AUTHOR TO CONFIRM]

3. Results

3.1 Group differences in cuneus GM volume and Picture Vocabulary

Table 2 reproduces the group comparisons exactly as printed on the poster, including the inequality notation. Full-term adolescents are the reference group.

Table 2. Group effects reported on the poster.

Outcome (model)	Group vs full-term	t	p, as printed
Cuneus GM volume (GM ~ group + sex + gestational age)	Moderately premature	4.36	p < 1.3e-5
Cuneus GM volume	Very premature	3.84	p < 1.2e-4
Picture Vocabulary (Cognition ~ group + sex + gestational age)	Moderately premature	−3.13	p = < 1.77e-3
Picture Vocabulary	Very premature	−5.59	p = < 2.45e-8

Observation. Cuneus GM volume was higher in both premature groups than in the full-term group. On the poster's plot, the very premature group's mean was the highest of the three. Picture Vocabulary scores were lower in both premature groups. Degrees of freedom, effect sizes and the definition of the plotted error bars are not reported [AUTHOR TO CONFIRM]. The poster also reports that five of the 100 GM regions showed significant group differences. Only the cuneus is identified.

3.2 Mediation

The poster's mediation diagram reports the path values a = 0.23 (gestational age → cuneus GM), b = −0.24 (cuneus GM → Picture Vocabulary), c = −1.33 (total effect) and c′ = −1.27 (direct effect). The poster states that the cuneus "mediated 4.1% of the association between gestational age and Picture Vocabulary performance." The poster does not report a test of the indirect effect, a confidence interval or whether the coefficients are standardized.

Observation. Cuneus GM volume statistically accounted for a small proportion of the association between gestational age and Picture Vocabulary score.

Two features of these values need author review, and this draft does not resolve either:

- **Proportion mediated.** From the rounded path values, a × b / c ≈ 4.15%, close to the reported 4.1%, while (c − c′) / c ≈ 4.5%. The estimator behind the reported figure should be stated.
- **Coefficient signs.** If gestational age is coded in weeks, the group results imply negative a and positive c. The reported a is positive and c is negative. This may reflect reversed coding of gestational age, a different variable from the one labelled, or a labelling error. [AUTHOR TO CONFIRM]

4. Discussion

This analysis found that adolescents born preterm had higher cuneus GM volume and lower Picture Vocabulary scores than full-term peers. Cuneus volume statistically mediated about 4% of the gestational age–vocabulary association. The preterm vocabulary gap is consistent with meta-analytic evidence of lower cognitive scores after preterm birth [3].

The direction of the GM finding differs from some prior work. Nosarti et al. reported *reduced* cuneus GM in very preterm adolescents, alongside increased GM in other regions [4]. Ma et al. reported that lower gestational age in ABCD was associated with *smaller* cortical volumes in a range of regions, which they described as fronto-parieto-temporal, fusiform, cingulate, insula and other areas [5]. Differences in method could contribute: component-based GM estimates versus atlas-based volumes, different gestational groupings and different covariates. Whether they explain the discrepancy has not been tested here (hypothesis).

The poster interprets higher cuneus GM as possibly reflecting a less mature developmental pattern. Cortical volume and thickness generally decrease across adolescence [11], so higher volume at a single age could be consistent with differences in the timing of maturation. It could equally reflect other processes, and higher volume does not by itself indicate either benefit or delayed maturation. A developmental trajectory cannot be shown without repeated measures. Altered maturation is therefore offered as a hypothesis.

The poster also suggests that altered cuneus development may help explain lower performance on visually guided tasks. The Picture Vocabulary Test uses pictures as response options, but it is a measure of receptive vocabulary, a language ability [9]. A link between a visual-cortex region and

vocabulary scores therefore does not show that the vocabulary gap is visual in origin. Testing that would require measures that separate visual processing from language. The poster proposes extending the analysis to other GM regions and to broader visual and attention networks.

5. Limitations

- The design is observational. Mediation estimated from data collected at one timepoint cannot establish temporal order or causation, and the indirect effect is small.
- Potential confounders are not reported as covariates. Examples include birth weight, multiple birth, neonatal complications and socioeconomic factors [AUTHOR TO CONFIRM].
- The very premature group is small (n = 152), and births under 28 weeks are not analysed separately.
- Five of 100 regions differed by group, but only the cuneus is reported. Other regions' results and any multiplicity correction are unconfirmed.
- Only one of seven cognitive tests is reported.
- Group boundaries are partly unconfirmed (Section 2.2), and the mediation values contain unresolved inconsistencies (Section 3.2).
- Findings describe group-level associations in a research sample. They do not support diagnosis or clinical prediction for individuals.

6. Conclusion

In ABCD data, adolescents born preterm showed higher cuneus GM volume and lower receptive vocabulary scores than full-term peers. Cuneus volume statistically accounted for a small share of the gestational age–vocabulary association. These associations motivate further work but do not establish a mechanism. The methods details and inconsistencies flagged in this draft need to be resolved first.

Acknowledgements

As printed on the poster: "This research was supported by grants R01-MH13 0595 and R34-DA061483 from the National Institutes of Health." Grant numbers [AUTHOR TO CONFIRM]. The ABCD Study's required acknowledgement text [AUTHOR TO ADD] [8].

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Editorial appendix: status and author checklist

This is a working draft for author review. It is not submitted, accepted or published, and it has not been peer reviewed. It was drafted from the poster with AI writing assistance for Arya Shah’s review. It awaits review by all coauthors, and the listing of coauthors does not imply their approval of this text. All results come from the poster. No analyses were rerun and no outside source substitutes for missing methods or results.

Before this draft can be circulated further, the authors need to confirm:

- Analytic N and group counts, including the path from the ABCD cohort to 3,135 and the meaning of “Sample Cohort-1”
- Age range and study timepoint of imaging and cognitive assessment
- Term thresholds: full-term cutoff, handling of 37–38 weeks and the source of gestational age
- ABCD data release and access date
- Imaging preprocessing and quality control

- Component extraction: moo-ICAr method, template, number of components, units and a method citation
- Site and family effects: site random effect specification and family or twin nesting
- Covariates, particularly whether age at assessment and/or gestational age were included alongside group
- Intracranial or total brain volume adjustment
- Multiplicity correction across 100 regions and the identity of the five significant regions
- Mediation specification, indirect-effect test and confidence interval; sign of a and c; 4.1% versus values computed from rounded paths
- p-value notation in Table 2 (“p <” and “p = <” as printed)
- Picture Vocabulary score type and results for the other six NIH Toolbox tests
- IRB/ethics statement and data availability under the ABCD data use agreement
- Author contributions
- Funding: grant numbers as printed (“R01-MH13 0595”) and ABCD acknowledgement text
- Citation for the poster’s description of cuneus function
- Review and approval by Dawn Jensen, Vince Calhoun and Jingyu Liu